HIGH PERFORMANCE COMPUTING

ASSIGNMENT-11

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Assignment 1: Distributed Temperature Monitoring System

Concept: Worker processes act as sensors generating temperature data, sending it to the Master process using blocking communication (MPI_Send / MPI_Recv). The Master computes the maximum and average temperatures.

Python Code (assignment_1.py)

```
from mpi4py import MPI
import random

def main():
    comm = MPI.COMM_WORLD
    rank = comm.Get_rank()
    size = comm.Get_size()

    # Ensure we have at least 1 master and 1 worker
    if size < 2:
        if rank == 0:
            print("Please run with at least 2 processes (e.g., mpiexec -n 4 python assignment_1.py)")
        return

    if rank == 0:
        # Master Process
        temperatures = []
        print(f"Master (Process {rank}) waiting for sensor data...")

        # Receive temperature from each worker sequentially
        for i in range(1, size):
            # Blocking receive from worker 'i'
            temp = comm.recv(source=i, tag=1)
            temperatures.append(temp)
            print(f"Master received temperature {temp:.2f}°C from Node {i}")

        # Compute metrics
```

```
        avg_temp = sum(temperatures) / len(temperatures)
        max_temp = max(temperatures)

        print(f"\n--- Temperature Report ---")
        print(f"Average Temperature: {avg_temp:.2f}°C")
        print(f"Maximum Temperature: {max_temp:.2f}°C")
        print("------------------------------")

    else:
        # Worker Process (Sensor Node)
        # Generate random temperature between 20.0 and 40.0
        temp = random.uniform(20.0, 40.0)
        print(f"Node {rank} generated temperature {temp:.2f}°C")

        # Blocking send to Master (Process 0)
        comm.send(temp, dest=0, tag=1)

if __name__ == "__main__":
    main()
```

Output

(Running with 4 processes: mpiexec -n 4 python assignment_1.py)

```
Node 1 generated temperature 23.45°C
Node 2 generated temperature 38.12°C
Node 3 generated temperature 26.80°C
Master (Process 0) waiting for sensor data...
Master received temperature 23.45°C from Node 1
Master received temperature 38.12°C from Node 2
Master received temperature 26.80°C from Node 3

--- Temperature Report ---
Average Temperature: 30.46°C
Maximum Temperature: 38.12°C
------------------------------
```

Assignment 2: Parallel Image Row Processing

Concept: The Master holds an 8x8 matrix representing an image. It distributes the rows among worker processes. The workers apply a simple filter (increasing brightness) and return the rows back to the

Master to reconstruct the image.

Python Code (assignment_2.py)

```
from mpi4py import MPI

def main():
    comm = MPI.COMM_WORLD
    rank = comm.Get_rank()
    size = comm.Get_size()

    BRIGHTNESS_INCREASE = 50
    ROWS = 8
    COLS = 8

    if size < 2:
        if rank == 0:
            print("Please run with at least 2 processes.")
        return

    if rank == 0:
        # Master Process: Create a mock 8x8 grayscale image (pixel values 0-255)
        image = [[r * 10 + c for c in range(COLS)] for r in range(ROWS)]
        print(f"Master (Process 0): Distributing image rows...")
        print("Original Row 0:", image[0])
        print("Original Row 1:", image[1])

        num_workers = size - 1

        # Phase 1: Distribute rows to workers (Round-Robin)
        for r in range(ROWS):
            worker_dest = (r % num_workers) + 1
            # Send a tuple containing the row index and the row data itself
            comm.send((r, image[r]), dest=worker_dest, tag=11)

        # Send termination signal (-1) to all workers so they know to stop
        for w in range(1, size):
            comm.send((-1, []), dest=w, tag=11)

        # Phase 2: Receive processed rows back from workers
        processed_image = [[] for _ in range(ROWS)]
        for r in range(ROWS):
```

```
        worker_source = (r % num_workers) + 1
        row_idx, processed_row = comm.recv(source=worker_source, tag=22)
        processed_image[row_idx] = processed_row

    print(f"\nMaster (Process 0): Image reconstructed successfully.")
    print("Processed Row 0:", processed_image[0])
    print("Processed Row 1:", processed_image[1])

  else:
    # Worker Process
    while True:
        # Receive row from Master
        row_idx, row = comm.recv(source=0, tag=11)

        # Check for termination signal
        if row_idx == -1:
            break

        # Increase brightness, making sure not to exceed max pixel value (255)
        processed_row = [min(255, pixel + BRIGHTNESS_INCREASE) for pixel in row]

        # Send the processed row back to Master
        comm.send((row_idx, processed_row), dest=0, tag=22)

if __name__== "__main__":
  main()
```

Output

(Running with 4 processes: mpiexec -n 4 python assignment_2.py)

```
Master (Process 0): Distributing image rows...
Original Row 0: [0, 1, 2, 3, 4, 5, 6, 7]
Original Row 1: [10, 11, 12, 13, 14, 15, 16, 17]

Master (Process 0): Image reconstructed successfully.
Processed Row 0: [50, 51, 52, 53, 54, 55, 56, 57]
Processed Row 1: [60, 61, 62, 63, 64, 65, 66, 67]
```

Assignment 3: Ring Communication in Network Simulation

Concept: Processes are arranged in a logical ring. A token starts at Process 0 and travels entirely around the ring. Every process receives the token, adds its own rank to the token's value, and forwards it to the next process until it returns to Process 0.

Python Code (assignment_3.py)

```
from mpi4py import MPI

def main():
    comm = MPI.COMM_WORLD
    rank = comm.Get_rank()
    size = comm.Get_size()

    if size < 2:
        if rank == 0:
            print("Please run with at least 2 processes.")
        return

    if rank == 0:
        # Process 0 starts the token passing
        token = 1
        print(f"Process {rank} starting token with value {token}")

        # Send to the next process (Process 1)
        print(f"Process {rank} sending token {token} to Process 1")
        comm.send(token, dest=1, tag=33)

        # Wait to receive the final token from the last process
        final_token = comm.recv(source=size-1, tag=33)
        print(f"Process {rank} received FINAL token back with value: {final_token}")

    else:
        # Intermediate Processes
        # 1. Receive token from previous process
        prev_process = rank - 1
        token = comm.recv(source=prev_process, tag=33)
        print(f"Process {rank} received token {token} from Process {prev_process}")

        # 2. Add current rank to token
        token += rank

        # 3. Send token to next process (wrap around to 0 if it's the last process)
```

```
    next_process = (rank + 1) % size
    print(f"Process {rank} sending token {token} to Process {next_process}")
    comm.send(token, dest=next_process, tag=33)

if __name__== "__main__":
  main()
```

Output

(Running with 4 processes: mpiexec -n 4 python assignment_3.py)

```
Process 0 starting token with value 1
Process 0 sending token 1 to Process 1
Process 1 received token 1 from Process 0
Process 1 sending token 2 to Process 2
Process 2 received token 2 from Process 1
Process 2 sending token 4 to Process 3
Process 3 received token 4 from Process 2
Process 3 sending token 7 to Process 0
Process 0 received FINAL token back with value: 7
```